

Sled for Cutting Staves on a Table Saw

Charles D. Hepler December 2014

Long compound angle staves may be used to make turning blanks for lamp standards, deep narrow vessels, etc. A sled is needed for cutting such staves because they must be very accurate if they are to fit together well as a turning blank.

The principles of cutting these staves are the same as cutting shorter compound miter segments. The staves must be cut exactly, to form tightly fitting cones. The requirement for high accuracy is ameliorated somewhat by the fact that any even number of staves can be assembled into half cones. The edges of the half cones should be coplanar. That is, they should rest solidly on a flat surface with no gaps. If they are not coplanar, they can be sanded on a flat surface until they are. Two half cones prepared in this way will fit perfectly. Long half cones, however, can be a bit difficult to sand perfectly flat. (See Chapter 42, *Sled for Cutting Segments*.)

The sled for long staves is somewhat different than the sled for compound segments. A compound *crosscut* miter sled won't work, because staves are ripped instead of crosscut. Often, a stave is cut from a billet that is rectangular in cross-section, although it is sometimes possible to cut more than one stave from a wider board.

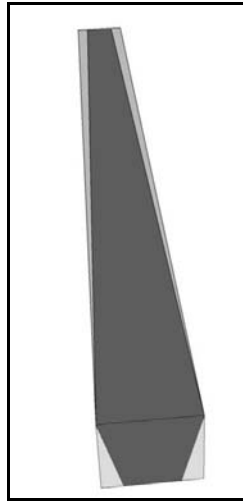


Figure 1. Example of Long Compound Angle Stave. The light gray shows the wood that will be cut off to make the stave.

Cutting the first side is straightforward, but this changes the reference edge, so the angle of the taper guide (ripping guide) has to change. To preserve the correct bevel angle the workpiece should be rotated horizontally. Therefore, to make the second cut, the miter angle has to change from a few degrees left of 0° to a few degrees right.

The typical commercial taper guide can't do that. Also, a typical taper guide can't hold a narrow piece securely. Holding a narrow stave with fingers would be foolhardy. A sled allows a hold down to clamp each piece in place to provide repeatably accurate cuts.

The sled(s) needed to make compound angle staves are trouble to make, so you might consider making straight (untapered) beveled staves on the table saw and then turning the assembled piece to the desired taper or other shape on the lathe.

Some people use angled router bits to do this operation, but a table saw seems much more versatile.

Geometry

The geometry involved is straightforward but may be confusing somehow at first. Let:

- L = the length of the stave.
- W = the width of the widest end of the taper
- N = the width of the narrowest end of the taper
- n = the width to remove from each side of the narrow end $([W-N]/2)$

(Both W and N must be measured on the same surface.)

The change in width over the length

of the stave is (W-N). Half of the change in width is on each side, so the angle is specified by rise over run (W-N)/2L. Use this ratio to make the taper guide. It also happens to be the tangent of the angle, so we can find the angle in degrees if we like (although we do not really need to know the angle in degrees).

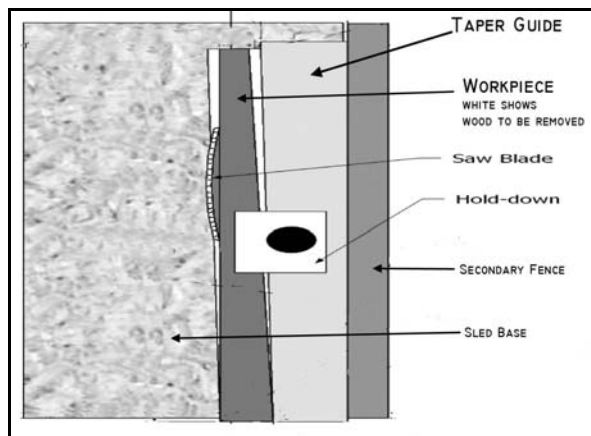


Figure 2. Sled Set Up for First Cut.

For example, suppose you want to cut staves that are 16½" long, 1-7/32" wide at their widest end, and 21/32" wide at the narrow end. The difference in width (W-N) is 9/16" and you need to cut off half of that (9/32") from each side of the narrow end. You can make a taper guide with this angle to run between the rip fence and the saw blade. (See Figure 2).

The taper per inch is

$$\begin{aligned} & (W-N)/2L \\ & ((1-7/32) - (21/32)) / 33 \\ & 9/16 / 33 \\ & .017" \end{aligned}$$

This number is also the tangent of the angle. The theoretical miter angle is 0.93° or .976° if you round off to the nearest 1/32°.

After the first cut, the wide end of the stave is still 1-7/32" and the narrow end is now 15/16". We have to cut 9/32" off the

opposite edge. ($15/16 - 21/32 = 9/32$). Because the saw blade is tilted we have to rotate the workpiece 180° horizontally to get the bevel in the desired direction. This puts the cut edge against the taper guide and the wide end of the stave at the back of the sled. (See Figure 3).

Because the reference edge of the stave is now tapered, we need a taper guide

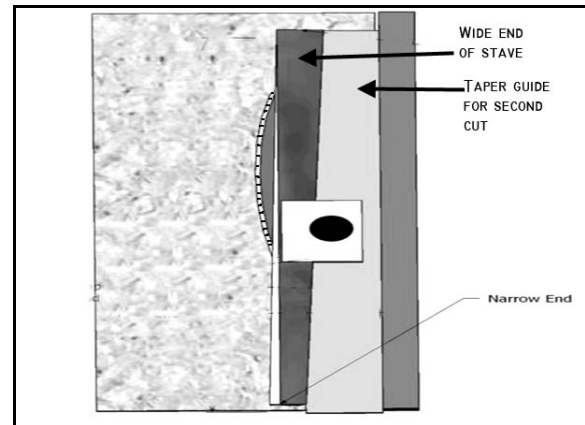


Figure 3. Sled set up for second cut.

with twice the angle as the first, i.e., 9/16" over 16½" or .034" per inch (1.95°).

The Basic Sled

The sled can be simply designed, much like any other saw sled, long enough to accommodate likely needs. It will run against the rip fence, so there is no need for rails to run in the miter slots. This example assumes a left-tilt blade, so the rip fence and the sled will run against the rip fence to the right of the blade

1. The sled should have its long sides parallel, so rip both long edges of the base on the table saw.
2. Securely attach (with glue and countersunk screws) a secondary fence along the right edge (which will rub against the rip fence).

3. The left edge of the sled can be sized to align with the saw blade (saw path). This works well enough, but support for the workpiece is then limited. It is better if the sled extends to the left beyond the saw blade, as shown in the figures, to better support the workpiece. In that case, you would need a “bridge” over both ends (not shown in the figures) to hold the two sides of the sled together after the kerf is cut. The kerf would eventually be widened by cutting staves of different widths.

4. The taper guide should be a bit thicker than the workpiece so that a beveled edge on the workpiece will not lift up when you press it against the guide. The hold down should prevent this, also..

5. The sled should provide a hold-down so that the workpiece does not ride up when pressed against the guide and so that the taper guide will move with the workpiece. The hold down should be sturdy because strong clamp pressure may be needed to hold the workpiece against the narrow shelf between the taper guide and the edge of the sled. A 5/16" threaded insert into a 1" thick hardwood guide seems strong enough, or the holddown can be attached to the pivot axle described below.

6. Attach a handle and stop block (workpiece stop) near the rear end of the sled (the end toward you). The handle should be such that you can slide the sled firmly against the rip fence without allowing your fingers to come within 3" of the saw blade at any moment during the cut. A piece of 2x4 on edge that spans the width of the sled from the saw path to the secondary fence worked OK. Screw it to the base. Keep the screws well away from the saw path. Mark a reminder on the handle about where to hold the sled and/or where not to put your fingers. The stop block is to keep

the stave from sliding while being cut. It should be located where the rear end of the stave will be when it is being cut.

Don't Put the Guide on the Left

Instead of using two taper guides on the right side of a left-tilting saw blade, some people just use a guide to the left of the blade to make the second cut. Moving the guide to the left of the saw blade seems easier, of course, but is an unreliable and potentially unsafe method for ripping a bevel because the saw blade is tilted into the work. The blade may bind and burn the workpiece, or even kick it back. (Note that when the blade is tilted away from the work the cutoff is free to fall away from the blade.) If you plan to sand the glue surfaces before joining the staves into half-cones, perhaps a small amount of burning would not be a problem, but kickback or overloading the saw motor should be a real concern.

Pivoting Taper Guide

As an alternative to making two taper guides, a pivoting taper guide can be used. I took this up as an interesting challenge, but you may not find it so. The idea is to make a relatively narrow (but stiff) taper guide that has an axle at the midline (front-to-back). To cut the taper, the guide pivots through an angle from a few degrees left to a few degrees right of zero (negative to positive, if you like).

Line 1. This is the guideline for the first cut (on a non-tapered billet). Make mark “A” W inches to the *right* of the saw track next to the stop block, where the rear end of the billet will be when cutting. (See Figure 4.) Mark L inches toward the front, parallel to the saw track. Turn a right angle and make mark “B” n inches $([W-N]/2)$ to the *right* from the saw

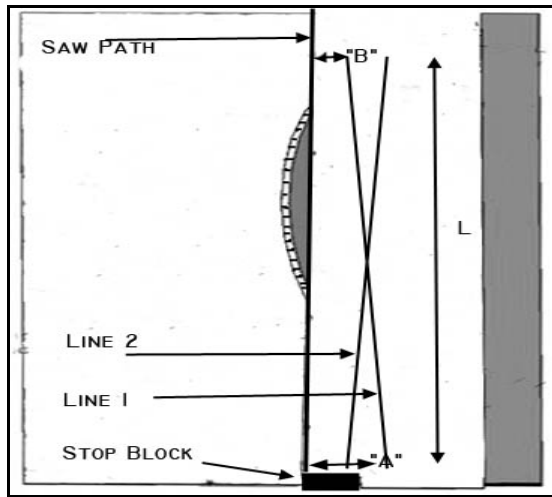


Figure 4. Diagram of Pivoting Taper Guide

track. (Measure W and n from the left side of the saw track or saw blade.) Draw a line between marks “A” and “B”. This will be a reference line, for the left edge of the taper guide, so make it permanent

2. Measure and mark the front-to-back midline of the fence ($L/2$ inches from the stop block.)

3. Set the left edge of the taper guide along line 1 and drill a hole for the axle through the taper guide and into the sled base. Attach the taper guide to the sled base with a wood screw or, preferably, a flat head machine screw from the bottom of the sled.

4. Reference Line. The saw path is the reference at zero degrees. Turn the taper guide parallel to the saw path and draw a line along its left side. This is the zero reference.

5. **Line 2.** This is the guideline for the second cut – after one side of the stave has been tapered. Measure W inches to the right of the saw path, L inches from the stop block

Measure N inches to the right of the saw path near the stop block. (again, measure from the left side of the saw path or saw blade.) Rotate the taper guide to coincide with these two marks and draw a line. See Figure 4 and Table 1.

The guide should pivot between the two layout lines. The edge of the guide does not have to be on the layout lines but it has to be exactly parallel to each of them. That is, it has to provide two accurate cutting angles.

To check, you can lay out the cut marks on the blank, as if you were going to cut it by hand. If you mark the workpiece and lay it against the saw track, the opposite edge will give you the position and angle of the taper guide for that cut.

Table 1. Layout Dimensions for Guidelines

	Guide Distance from Saw Line		
	Front of Sled (Top of Figure 3)	Back of Sled (Bottom of Figure 3)	Tangent (Ratio to lay out guideline)
Line 1	$\frac{1}{2}(W-N)$	W	$(W+N)/(2L)$
Line 2	W	N	$(N-W)/L$

Adjustment Blocks (Spacers)

Cut small blocks of wood to fit between the guide and the secondary fence. The spacers do not need to be any particular width, but the maximum width is the distance from the rear of the guide to the secondary fence (See Figures 5 and 6.)

Wedge these blocks between the guide and the secondary fence. Draw a mark across the spacer and the secondary fence and label it.

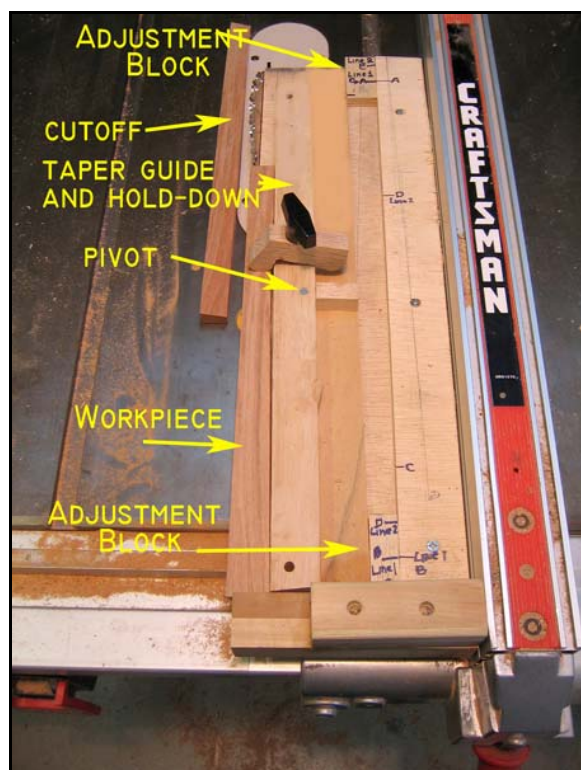


Figure 5. Sled in Position for First Cut. Note position of adjustment blocks

The spacers and the pivot screw will hold the guide securely in position and allow you to quickly reset the position of the pivoting guide to make precisely repeatable cuts.

Cutting the Staves

Some authorities recommend that beveled tapers be cut in two steps: taper first, then bevel. Some one-step procedures can burn the work if the saw is slightly mis-adjusted or if the workpiece is able to bind the saw blade during the cut.

The sled described here can cut beveled tapers in one step, at least on my table saw. The two-step procedure uses wood more efficiently if more than one stave can be cut from one wider board. In the example we have been using, two $1\frac{7}{32}$ " staves require 2" from

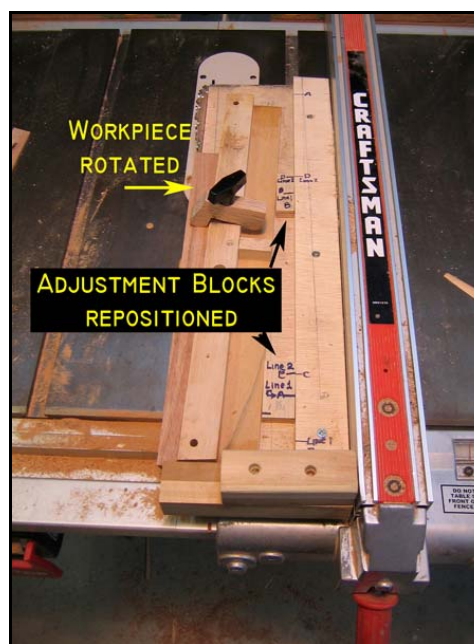


Figure 6. Sled in Position for Second Cut. Note new position of adjustment blocks.

a wide stock board with a two-step process (allowing $\frac{1}{8}$ " for kerf.). Using a one-step process, two staves require $2\frac{9}{16}$ ", more than 25% more. That's because cutting the tapers before the bevels allows the tapers to be cut in an alternating sequence that leaves less cutoff (waste) stock.

So, the one-step process saves time but may waste wood if the stock lumber is in wide boards. The two-step process takes longer (and requires two sleds or three taper guides) but can save wood.

ONE-STEP PROCEDURE

The one-step procedure begins with rectangular billets of length L and width W or wider (but not wider than $W+N+\text{saw kerf}$). To cut the staves in one step, tilt the blade to the desired angle (e.g., 14.97° left of vertical or 75.03° right of horizontal).

I use a digital tilt gauge. A triangular

gauge cut out of plywood or a sliding bevel gauge (laid out with trigonometry) should be a bit more accurate, but in reality it is a bit difficult to see the gap between the saw blade and the gauge.

Setting the sled for width of cut W is a bit tricky because the blade is tilted. Probably the easiest way is to lay out the cut lines, including the bevel angle, on one of the billets and then set the sled (in position 1) to cut those lines. After the first cut, move the guide to position 2, rotate the workpiece 180° horizontally, and cut the second side. Don't flip it over. The wide side stays "up." Check

that the bevel angle will be in the right direction. You need a *trapezoidal* cross section, not a parallelogram.

You will again cut off $\frac{1}{2}(W-N)$ inches from the other side of the narrower end, but now the narrower end will be toward the front of the saw. (See Figures)

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